

Program3

Code a dockerized python flask or node.js application

step 1: create folder

>> mkdir program3

>> cd program3

```
isha@meow:~/DevOps$ mkdir program3
isha@meow:~/DevOps$ cd program3
isha@meow:~/DevOps/program3$ nano app.py
isha@meow:~/DevOps/program3$ nano Dockerfile
isha@meow:~/DevOps/program3$ nano requirements.txt
isha@meow:~/DevOps/program3$
```

step2: create files inside this folder

>> nano app.py

```
GNU nano 7.2 app.py *
from flask import Flask

app = Flask(__name__)

@app.route('/')
def hello_world():
    return '<h1>Hello from Docker!</h1><p>Your Flask application is running suc>

if __name__ == '__main__':
    app.run(debug=True, host='0.0.0.0', port=5000)
```

>>nano requirements.txt

```
GNU nano 7.2 requirements.txt
Flask==3.0.3
```

>>nano Dockerfile

```
GNU nano 7.2 Dockerfile
FROM python:3.9-slim

# Set the working directory inside the container
WORKDIR /app

# Copy the requirements file into the container
COPY requirements.txt .

# Install the needed packages
RUN pip install --no-cache-dir -r requirements.txt

# Copy the rest of your application code
COPY . .

# Expose the port Flask runs on
EXPOSE 5000

# Run the application
CMD ["python", "app.py"]

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

step3: build the dockerfile

>> docker build -t program3 .

```
i.sha@neow:~/DevOps/program3$ docker build -t program3 .
[+] Building 12.5s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 473B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
=> [internal] load build context
=> => transferring context: 825B
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:f672889c53fbf9e16d76aae429d1798614f09d506b536cc00c87dcc2c30bc0b7
=> => naming to docker.io/library/program3
i.sha@neow:~/DevOps/program3$
```

docker:default
0.0s
0.0s
3.8s
0.0s
0.0s
0.0s
0.0s
0.0s
0.0s
0.0s
8.2s
0.0s
0.3s
0.2s
0.0s
0.0s

Step 4: Run the Dockerfile and expose to the port and open the browser to see the message from the Docker.

>> docker run -p 5000:5000 program3

```
isha@meow:~/DevOps/program3$ docker run -p 5000:5000 program3
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 563-833-816
```

